

Population–Traffic Driven EV Charging Infrastructure Analysis using DBSCAN

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Abstract

In this study EV(Electric Vehicle) charging infrastructure across cities in the United States the balance between charging station availability, population demand, and traffic congestion. An underservice score was developed to identify cities where existing charging infrastructure may not adequately meet local needs, allowing cities to be ranked according to their level of infrastructure deficiency. To determine whether these underserved cities represent isolated cases or broader geographic patterns, DBSCAN spatial clustering was applied using city locations. The analysis identified 14 underserved regions, with approximately 95 percent of underserved cities belonging to regional clusters. This suggests that gaps in EV charging infrastructure are focused on high-density locations and not just randomly spread throughout regions. The differential demand and congestion factors across various regions demonstrate that the shortage of charging infrastructure results from differing conditions at local levels, highlighting the significance of local demand as well as regional patterns for planning future EV Charging Infrastructure.

I. INTRODUCTION

A rapid increase in EVs has led to numerous challenges related to charging infrastructure access and use in both urban and semi-urban environments. These challenges can be tackled by using a demand–supply Gap Analysis. This research uses data science to use guesswork in infrastructure planning by offering a more structured data driven method to identify locations where EV charging stations are actually needed. This work implements an end-to-end pipeline for clustering and demand–supply gap analysis. Real-world demand indicators like population density, traffic congestion levels and charging infrastructure requirements are incorporated. This project contains spatial data visualization, data integration from multiple sources as well as feature engineering from spatial data, understanding of clustering methods and structures of clusters, and multi-criteria decision making for evaluating and ranking locations. The study scope is limited to specific regions presenting EV adoption and mobility related data. This research focuses on space-based demand analysis identifying underserved areas for EV charging infrastructure. This proposed approach can be extended to other cities/regions having similar datasets.

II. PROBLEM STATEMENT

A critical challenge in planning for EV infrastructure is to identify where there is a lack of charging infrastructure compared to the demand and if there are deficiencies that exist as isolated problems at the city-level or larger regional patterns. A lack of systematic identification of underserved locations can lead to infrastructure planning and business investment decision making that does not mitigate locations with the greatest need.

Research Question: How can spatial clustering techniques applicable to geospatial data contribute to gap analysis in identifying underserve EV charging regions?

This research tackles this challenge by providing a framework for underserve EV assessment based on demand pressure and congestion pressure, while conducting spatial clustering techniques on regional hot spot areas. The goal is to be able to understand the intensity and spatial distribution of EV charging infrastructure gaps and help support better, data-based infrastructure planning decisions

III. RELATED WORK

Geospatial data is any data that has a location attached to it — a coordinate, address, or boundary — that tells you where something happened. It carries geographic, temporal, and attribute information together, meaning it captures where, when, and what all at once DBSCAN is the most practical choice for geospatial work because real-world spatial patterns like traffic bottlenecks or crime hotspots are irregular in shape, and DBSCAN handles that naturally while flagging outliers as noise.. [1])

This study analyzes the urban demand patterns for urban infrastructure planning using the DBSCAN-DBSCAN clustering method. This study shows that adding multiple spatial layers to the clustering data uncovers the hidden patterns. However, many existing approaches rely on rule-based spatial aggregation or predefined zoning methods, which are not an effective methods. [2], [3].

In this research, we look at how well supply/demand relationships between urban public service facilities and a city's population match up. These mismatches between the supply and demand of facilities within an urban area are very frequent. An additional conclusion from this research was that using accessibility as an evaluation tool for infrastructure helps determine the value and equity of public facilities based on how far away the facility is from the community it serves and how well services can be reached from those locations. [4]

It has been concluded that contemporary urban planning strategies need to be flexible, multi-layered and founded on actual spatial data rather than fixed assumptions. This supports your demand/supply optimization framework as the primary reason for making an adaptation to current planning methodologies. [5]

EVCS recently conducted studies that demonstrate the "need for data-driven scalable EV charging location optimization (compared to static planning)" as well as a higher level of heterogeneity in charging demand and greater impact from spatial and contextual elements including urban density, highway proximity and proximity to points of interest. Geographically weighted machine learning models are able to accurately represent dynamic patterns at the local level for EVCS charging demand, outperforming global models. [6]

DBSCAN handles noise and outliers well, labeling sparse regions as noise rather than forcing them into clusters. This property is especially useful in geo-spatial datasets where irregularities and sparse demand points are common. Unlike partitioning methods such as K-means, DBSCAN does not require the number of clusters to be predefined. Traditional methods of planning are mostly based on manual decisions (heuristics) results are based on rules of thumb like placing EVs in commercial zones and near the high populated areas. In our work we are depending on the data insights to optimise the locations which meets the demand-supply criteria. [7]

However, these improvements, many infrastructure planning methods require a heuristic approach where the number of battery chargers is determined using set criteria-based locations, such as distance (usually proximity) from commercial areas or dense population centers. This study utilized a geospatial gap analysis tool that looks at demand indicators (population and traffic congestion) and supply indicators (currently available public charging stations) to determine how many additional stations are needed in under-served areas to meet the needs of people using electric vehicles (EVs).

IV. METHODOLOGY/APPROACH

This research presents a data-based framework to detect underserved EV charging regions based on the combination of the indicators for demand–supply imbalance along with spatial clustering method.

A. Constructing Demand and Congestion Pressure

$$\text{Demand Pressure} = \log \left(1 + \frac{\text{Population}}{\text{Charging Stations}} \right) \quad (1)$$

$$\text{Congestion Pressure} = \frac{\text{Total Congestion}}{\text{Charging Stations}} \quad (2)$$

This framework is comprised of four major steps, namely pressure indicator construction, standardization, calculation of underservice index, and finally spatial clustering. The congestion pressure refers to the level of traffic around the charging facilities. High levels of this index show the location of charging stations within high traffic areas.

Congestion Pressure refers to the density of traffic around charging facilities. High levels denote that charging facilities are situated in highly trafficked areas, which will put more pressure on demand and make accessibility difficult.

B. Z-Score Standardisation

Since demand pressure and congestion pressure are measured on different scales, they cannot be directly combined. Z-score standardisation is applied:

$$z = \frac{x - \mu}{\sigma} \quad (3)$$

This converts both variables into a common comparable scale.

Interpretation of z-scores:

- Negative values → below-average pressure (better served cities)
- Zero → average conditions
- Positive values → above-average pressure (potentially underserved cities)

This ensures fairness when combining variables into a single index.

C. Underservice Score

$$\text{Underservice Score} = 0.6z_{\text{demand}} + 0.4z_{\text{congestion}} \quad (4)$$

Underservice Score Interpretation:

TABLE I: Interpretation of Underservice Score

Underservice Score	Interpretation
< 0	Relatively well served
≈ 0	Average infrastructure adequacy
> 0	More underserved than average
$\gg 0$	Highly underserved

An underserved city refers to a situation where there is an above average level of demand pressure, congestion pressure or even both. Such cities might have inadequate infrastructure facilities to cater to the demands of the population and transport activities taking place in the city.

This weighted index gives priority to population-induced demand and accounts for mobility pressure, providing a continuous scale of severity of the underserved infrastructure.

D. Spatial Clustering Analysis

The DBSCAN algorithm (Density-Based Spatial Clustering of Applications with Noise) helps detect geographically close clusters of cities that do not have sufficient services. Ranking methods allow you to see which cities actually have less service than others but do not work well because rankings treat each city on its own without taking geographic ties into account. Using the DBSCAN algorithm to find clusters will allow you to see which regions are underserved (or over-served), so that cities can plan better for their region's needs.

E. DBSCAN Implementation

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is applied to the underserved cities. The algorithm groups cities based on spatial proximity, where a city is assigned to a cluster if it has at least a minimum number of neighbouring underserved cities within a specified radius (eps). In this study, eps is converted into real-world distance using Earth's radius, allowing clustering to be interpreted in kilometers. Cities that do not satisfy the density requirement are treated as noise points and represent isolated underserved locations.

In the next step DBSCAN clustering was applied latitude and longitude of the cities. DBSCAN clusters by grouping cities that lie within a specified neighborhood distance of one another. This distance is controlled by the parameter ε (eps). A city is assigned to a cluster if it has at least 5 neighboring underserved cities within the specified geographic radius. DBSCAN to compute distances between cities and identify spatial clusters of underserved regions using geographical coordinates.

To convert the epsilon value into kilometers, the following approximation is used:

$$\text{Distance (km)} = \varepsilon \times 6371$$

where 6371 km is the approximate radius of the Earth.

For example:

$$0.02 \times 6371 \approx 127.4 \text{ km}$$

Thus, two underserved cities are considered neighbors if they are within approximately 127 km of each other.

A city must therefore have at least 5 underserved cities within this radius to be included in a cluster.

F. DBSCAN Evaluation

The clustering yielded 13 spatial clusters with noise ratio as 4.93 percent and coverage ratio as 95.07 percent. This indicates that most of the underserved cities have been clustered into regions, whereas very few of them have been left out as noise. The value of Silhouette score as 0.178 indicates moderate separation among clusters, which is obvious in spatial data where natural boundaries exist among different regions. Besides Silhouette Score, various other criteria for evaluating clusters were also used for determining the validity of clusters.

V. DATA DESCRIPTION

The primary dataset used in this study is obtained from the Alternative Fuels Data Center (AFDC) provided by the U.S. Department of Energy [8]. The EV charging station dataset contains thousands of charging station records across the United States with 84539 stations including public and private. The dataset includes multiple attributes covering station identity, location, accessibility, and technical specifications. In addition to this we have traffic and population data which are external datasets to perform spatial demand-supply analysis. For geospatial gap analysis and DBSCAN clustering, the most relevant features from the EV charging station dataset include latitude and longitude coordinates, which are essential for identifying spatial distribution.

Population dataset [9] is collected from “City Population Data (CSV)” source, which provides city-level population statistics across the United States the population estimates as of July 1, 2025.

Traffic data [10] is sourced from the TomTom Traffic Index for the United States, which provides information on congestion levels and Time lost during rush hour across 91 cities. There is no official API to collect the data data is collected using webscraping methods using playwright, requests and beautifulsoup python libraries.

All datasets were first stored locally and then cleaned to make them easier to work with and more consistent. Unnecessary columns were removed to reduce noise and focus only on relevant information. Missing values were handled in different ways depending on the type of data: numeric fields related to EV chargers were filled with zero. Different approaches were employed to handle missing data based on the type of field, e.g., where a numeric EV charger field had a missing value it was assigned a value of zero, whereas the categorical field (i.e., maximum vehicle class) had missing values replaced with 'unknown'. For standardising ZIP codes all were converted to a common integer format, and street addresses that were missing were imputed using GeoPy reverse geocoding. After cleaning data number of rows reduced to 78,954 records and the features to 25 which are necessary. City name consistency was creatively worked on first among all data sets to determine where joins would take place. The EV charging dataset was then merged into the population dataset, incorporating the forecast population in 2025. Cleaned traffic congestion percentages were converted to numbers (i.e., no percentage sign) and added to the combined dataset. The final combined dataset provides a comprehensive view of EV infrastructure, population and traffic to enhance opportunity for analyses based on demand/supply patterns.

VI. DATA ANALYSIS

A. Exploratory Data Analysis

The distribution of EV charging infrastructure across the U.S. is highly uneven, with a strong concentration in coastal and highly urbanized states for example California.

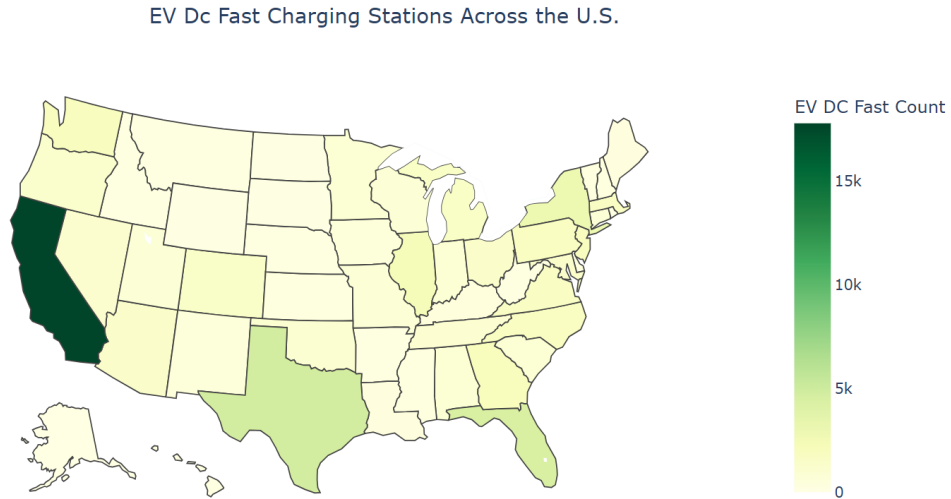


Fig. 1: DC fast charger distribution

Charging infrastructure for light duty vehicles is mostly located in major metropolitan areas, whereas rural area with lower coverage.

Level 2 chargers are widespread but slower charging access, DC fast chargers are limited and unevenly distributed, creating gaps in long-distance and high-demand travel corridors.

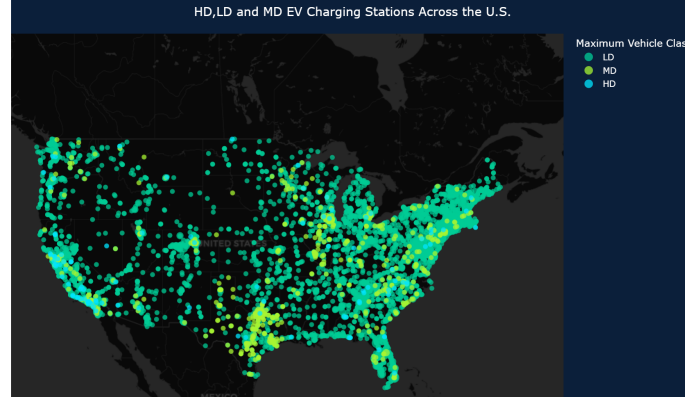


Fig. 2: Vehicle type distribution

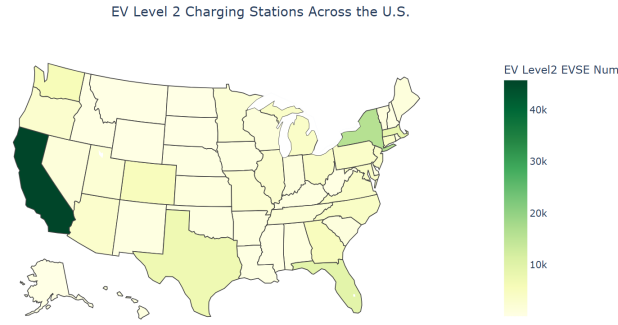


Fig. 3: Level 2 chargers distribution

B. City Ranking

Ranking of cities was done based on an underservice measure and incorporated into the overall dataset for priority location identification. From the results, it can be observed that the underserved cities are not randomly located but have regional clusters with some being isolated. It implies that the EV stations under service is not due to isolated factors but to regional factors. Nearby cities face similar problems and thus there must be some underlying factor causing this trend. The map has shown a non-uniform distribution of the EV stations where they are more concentrated in urban regions while few are available in the rural areas. The clustering of the data makes the application of DBSCAN appropriate for grouping of data in the geographic space.

C. DBSCAN

The parameter, Epsilon, was set at 0.02 using Haversine distance metric, which implies that the approximate radius of the neighborhood is 127 km. Using this parameter, cities were said to be in spatial proximity if they were in the 127 km distance. DBSCAN would therefore use this algorithm to group connected underserved cities in regional clusters where the clusters would show geographically concentrated infrastructure underserved areas. There are 13 geographically connected underserved clusters identified by DBSCAN. The biggest cluster is comprised of multiple neighboring states, implying a large regional concentration of underserved cities. Small clusters are hot spot regions. The extent of the clusters was not taken to be the exact geographic region since DBSCAN connects cities in a cluster rather than providing continuous coverage regions.

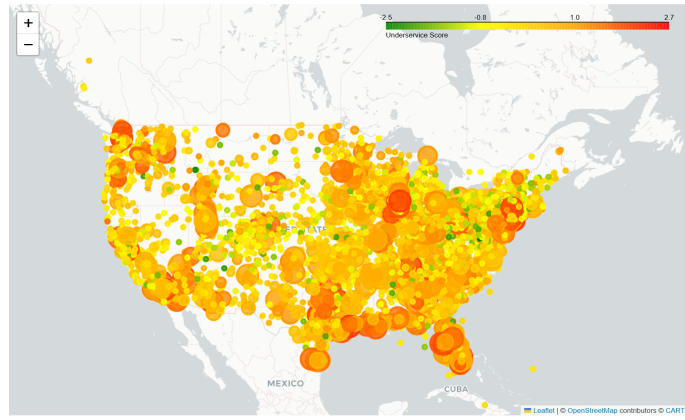


Fig. 4: Ranked cities

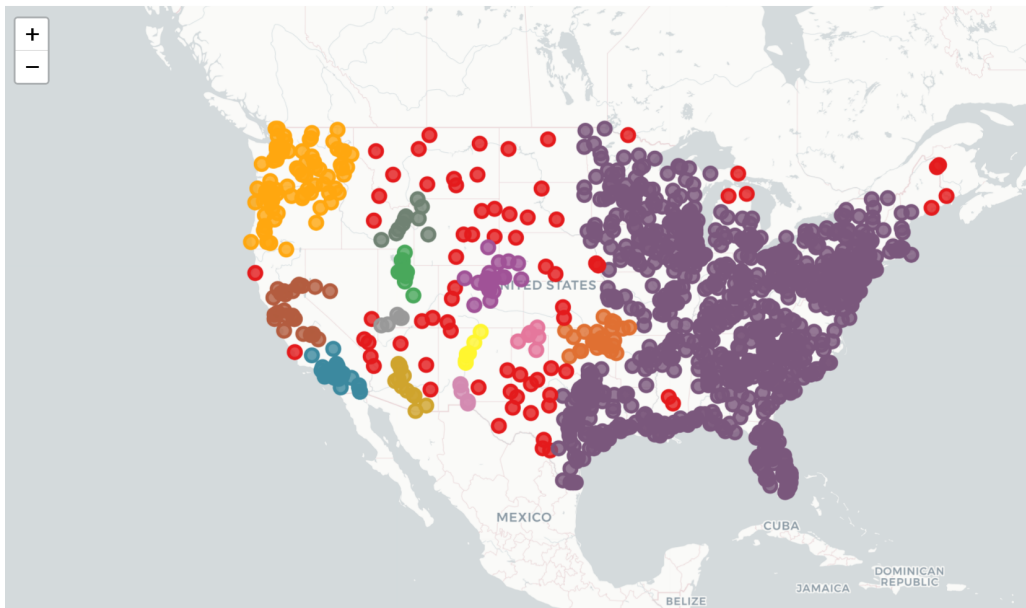


Fig. 5: DBScan Clusters

D. Model Tuning

Epsilon (eps) was tuned iteratively to identify a balance between over-fragmentation and over-merging of underserved regions. Smaller epsilon values (0.01–0.015) produced a large number of clusters (25–46) and classified many cities as noise, indicating that nearby underserved areas were being split into multiple small clusters. As the epsilon value was increased, the fragmented data was decreased by joining together individual hotspots into larger clusters/regions. An epsilon value of 0.034 was chosen as it resulted in only 13 distinct clusters with 73 noise points and an increased ability to represent clearly the regional underserved EV charging zones without losing any of the spatial characteristics associated with them.

E. Model performance

The model was able to generate 13 clusters, which means that the model has been able to find a fairly detailed segregation of underserved areas. There was just 4.93 percent of the cities that were considered to be noise, which means that there were very few points that did not belong to any cluster, and the clustering coverage of the cities is 95.07 percent, which means that most cities have been clustered into some hotspots.

TABLE II: DBSCAN parameter tuning

eps	Clusters	Noise Points
0.010	46	414
0.012	38	288
0.015	25	148
0.018	18	91
0.020	13	73

The silhouette score of 0.178 is relatively low, which means that the clusters are poorly segregated from each other. It is pretty much common with geographic clustering since regions tend to blend into one another.

TABLE III: Clustering performance summary for underserved regions

Metric	Value	Interpretation
Number of clusters	13	Identified 13 underserved regions
Noise percentage	4.93%	Very few cities remained unclustered
Coverage percentage	95.07%	Almost all underserved cities belong to a hotspot region
Silhouette score	0.178	Moderate separation, expected for geographic clusters

F. Spatial clustering

DBSCAN clusters are clusters of underserved cities which are grouped based on their proximity to one another through the presence of neighboring cities in close proximity. It should thus be noted that clusters show concentration of gaps rather than actual geographic service coverage areas.

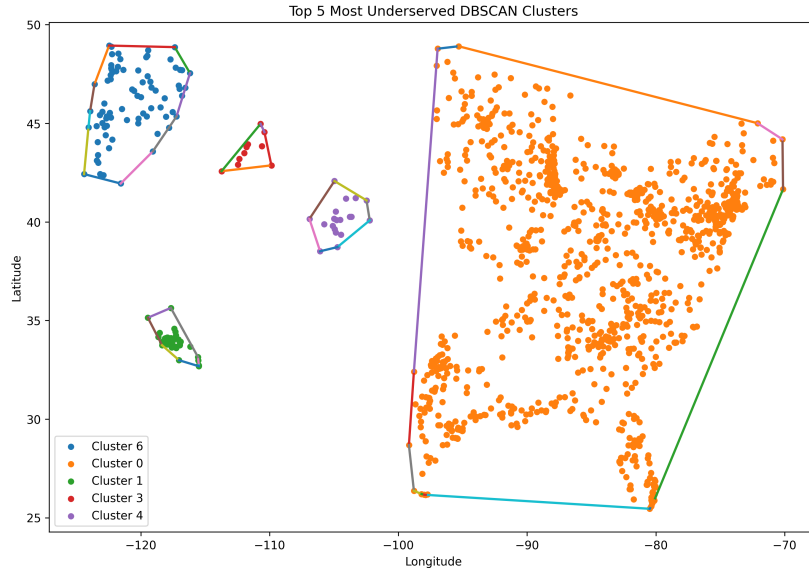


Fig. 6: Top 5 cluster boundaries

G. Infrastructure analysis

With regard to demand and congestion pressure, the first priority when it comes to planning will be Cluster 1 due to its high population density per station along with congestion pressure. The second priority will be Clusters 3 and 4 due to higher congestion and worse performance, which indicates that these clusters will need congestion reduction and optimization. Cluster 0 is larger and more moderate in nature; therefore, it will be considered a monitored cluster with gradual improvement. Cluster 6, on the other hand, has the best performance and efficiency and will not be a focus.

TABLE IV: Infrastructure indicators across clusters

Cluster	City Count	Avg Score	Pop/Station	Congestion	Max Score
6	78	0.9805	4292.98	35.56	1.9715
0	1151	0.9130	7623.06	26.40	2.7396
1	44	0.8731	13728.60	17.59	2.5658
3	10	0.8652	6891.12	28.12	1.7063
4	19	0.8213	5223.16	29.01	1.7253

VII. CONCLUSION

This paper proposes a data-driven framework for characterizing underserved regions for EV charging. By adding the demand–supply imbalance metrics with spatial clustering methods, this study finds that infrastructure underservice can be quantified with a singular underservice score constructed from both demand pressure and congestion pressure metrics; cities with similar levels of underservice can be grouped consistently; cities with high underservice do not randomly scatter on the map but cluster in clear regions, suggesting that underlying factors that are spatially correlated may also contribute to infrastructure underservice. DBSCAN further separates regionally-cited hotspots from outliers. This highlights how spatial methods go beyond basic ranking systems. This foundational framework can be expanded by including demographic factors like income, vehicle ownership, city density, etc., accounting for regional policy variations, as well as temporal metrics to capture the dynamics of infrastructure gaps over time. This work ultimately aims to create a scalable and interpretable framework for informing data-driven decisions to plan EV charging infrastructure.

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